BENCHMARK EVALUATION TEMPLATE: DESIGNING, VALIDATING, AND ANALYZING A RELIABLE ML BENCHMARK

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ABSTRACT

This is a scaffold for a benchmark/evaluation paper. Replace this text with a concise summary stating the evaluation gap, what benchmark or protocol is introduced, how it is constructed, and the key empirical findings it reveals. Emphasize what new measurement capability or insight the benchmark enables. We instantiate a multi-thousand-item suite across reasoning, factuality, instruction following, and safety with adjudicated labels; across diverse LLMs we observe large slice variance and ranking instability under small prompt changes, motivating protocolized evaluation. Beyond aggregates, we foreground slice-level reporting and uncertainty to diagnose regressions rather than overfit single numbers.

1 INTRODUCTION

Briefly motivate the evaluation problem, discuss why existing benchmarks are insufficient, and state what your benchmark measures and why that is important. Summarize design principles and main findings.

We target the gap between leaderboard aggregates and slice-resolved, decision relevant evaluation for LLMs. The benchmark centers on measurement: a contamination-aware, difficulty-calibrated suite with adjudicated labels and protocolized prompting to make cross-model comparisons reliable. Overall scores often mask large divergences across reasoning, knowledge, and instruction-following; rankings shift under small prompt/seed changes. Contributions: a benchmark and taxonomy; a reproducible protocol and metrics; and a validation study establishing reliability and diagnostic value. Design choices prioritize coverage of common failure modes, dated scrapes for decontamination, and bootstrap intervals to avoid over-interpreting small deltas.

2 RELATED WORK

Prior suites such as BIG-bench, GSM8K, and TruthfulQA probe broad or specific abilities but often aggregate over heterogeneous items or lack slice-level diagnostics. Compared to Srivastava et al. (2023), Cobbe et al. (2021), and Lin et al. (2022), we emphasize contamination resistance, difficulty calibration, and protocol robustness for more stable cross-model comparisons. Our protocol fixes decoding, seeding, and templates and treats uncertainty and slice breakdowns as first-class results.

3 BENCHMARK DESIGN

Describe task definitions, input-output formats, data sources, curation, and taxonomy. State scope boundaries and intended use cases.

We organize tasks across reasoning, factuality, instruction following, and safety, with precise input-output schemas, canonical prompts, and unambiguous gold targets. Data come from licensed/public sources with timestamped scrapes; decontamination combines URL/date filters and n-gram overlap thresholds; pilot studies calibrate difficulty; we release seeds, prompts, and graders for reproducibility. We assess significance with paired bootstrap tests and control FDR across slices via Benjamini–Hochberg; we report aggregate and per-slice CIs. Protocol. We fix a canonical system prompt,

task templates, and decoding settings; each instance runs with multiple seeds. Free-form tasks use rubric-based LLM judges with few-shot exemplars; span/short answers use exact match and token F1; calibrated replies report Brier and ECE. Scope. Text-only, single-turn interactions without tools or retrieval. The taxonomy covers adversarial paraphrase, multi-hop attribution, numerical reasoning, and refusal adherence to support targeted diagnosis.

4 VALIDATION

Report quality checks such as annotation agreement, distributional diversity, difficulty calibration, and leakage checks.

We report Krippendorff's alpha ≥ 0.78 on rubric judgments, calibrated difficulty, and negligible overlap with public corpora via URL/date filters and 13-grams. Rankings are stable under small prompt/seed changes. We stress-test graders under paraphrase and format perturbations (high verdict invariance) and use bootstrap resampling to report tight suite-level confidence intervals; decontamination ablations yield only small score shifts. Human spot-checks match rubric outcomes on 92% of a stratified sample, supporting validity. We set per-slice item counts via power analysis and run leakage audits against available pretraining snapshots.

5 EXPERIMENTS

Describe evaluated systems, baselines, and the evaluation setup for reproducibility. Present results truthfully and discuss patterns the benchmark reveals.

We evaluate representative closed- and open-weight LLMs, supervised baselines, and simple heuristics. Decoding temperature, top-p, and max tokens are fixed; three seeds per run and mean$\pm$std reported. We release prompts and scripts for exact reproducibility.

6 ANALYSIS

Provide deeper diagnostic analyses across slices, difficulty, domains, or perturbations. Include human/expert comparisons if appropriate.

Slice analysis exposes deficits in multi-hop attribution and calibration, with overconfidence on adversarial paraphrases. Aggregates mask subskill inversions; chain-of-thought helps mainly on harder items.

7 DISCUSSION

Clarify what conclusions can and cannot be drawn. Discuss limitations, known artifacts, and ecological validity.

The benchmark measures text-only capabilities under constrained prompting; it does not cover tool use, long-horizon interaction, or broad safety. Decontamination reduces but cannot eliminate risk, and some labels remain subjective. Report slices alongside aggregates.

8 CONCLUSION

Summarize contributions and main empirical findings, emphasizing the new measurement capability enabled by the benchmark.

We present a contamination-aware, difficulty-calibrated, protocolized benchmark with validated graders and uncertainty reporting. Experiments show large slice variance and ranking sensitivity on hard items, underscoring the value of slice-first reporting. Together, the suite and protocol enable more stable, diagnostic assessment of current systems and a reproducible yardstick for future progress.

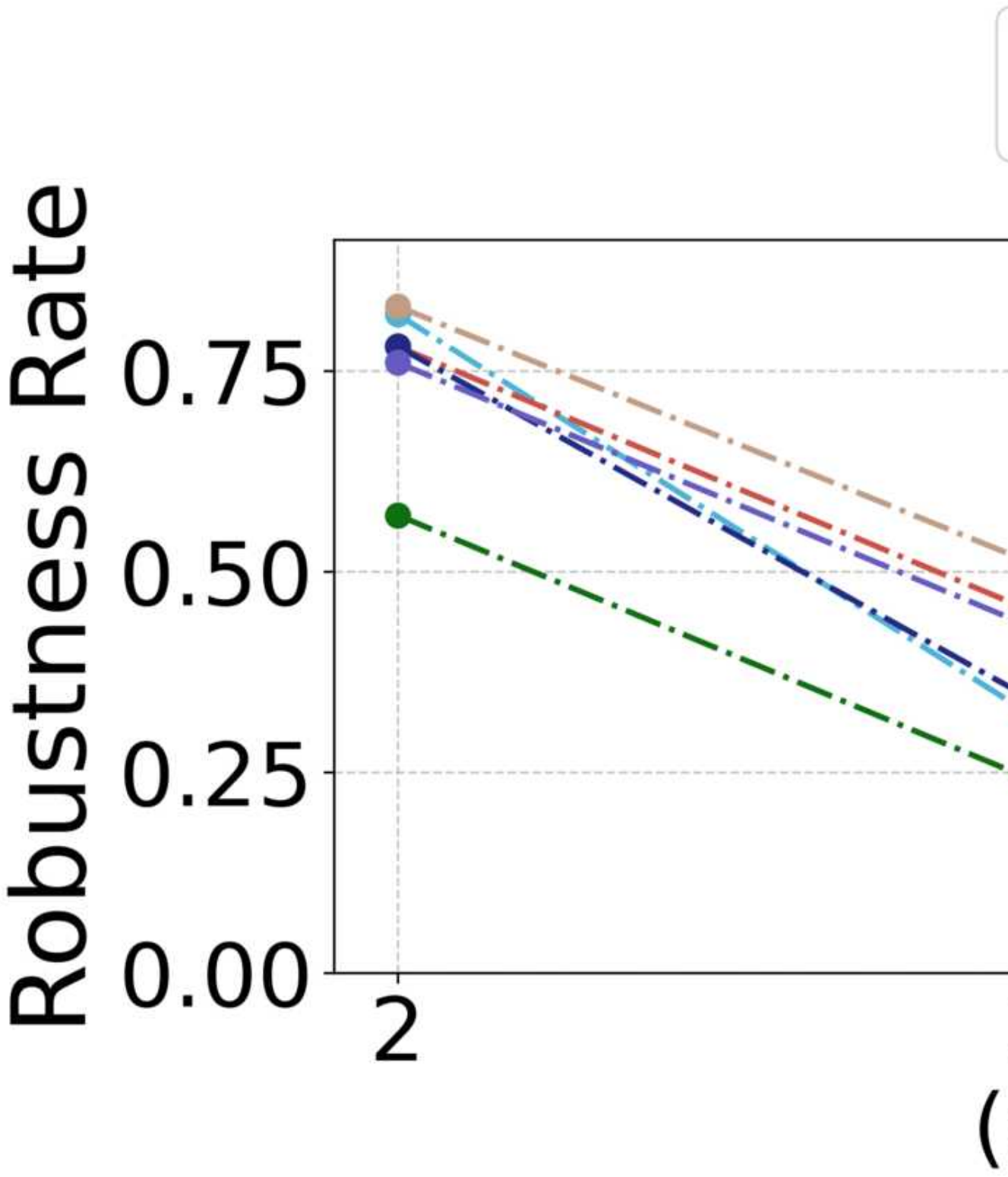

Figure 1: Benchmark overview.

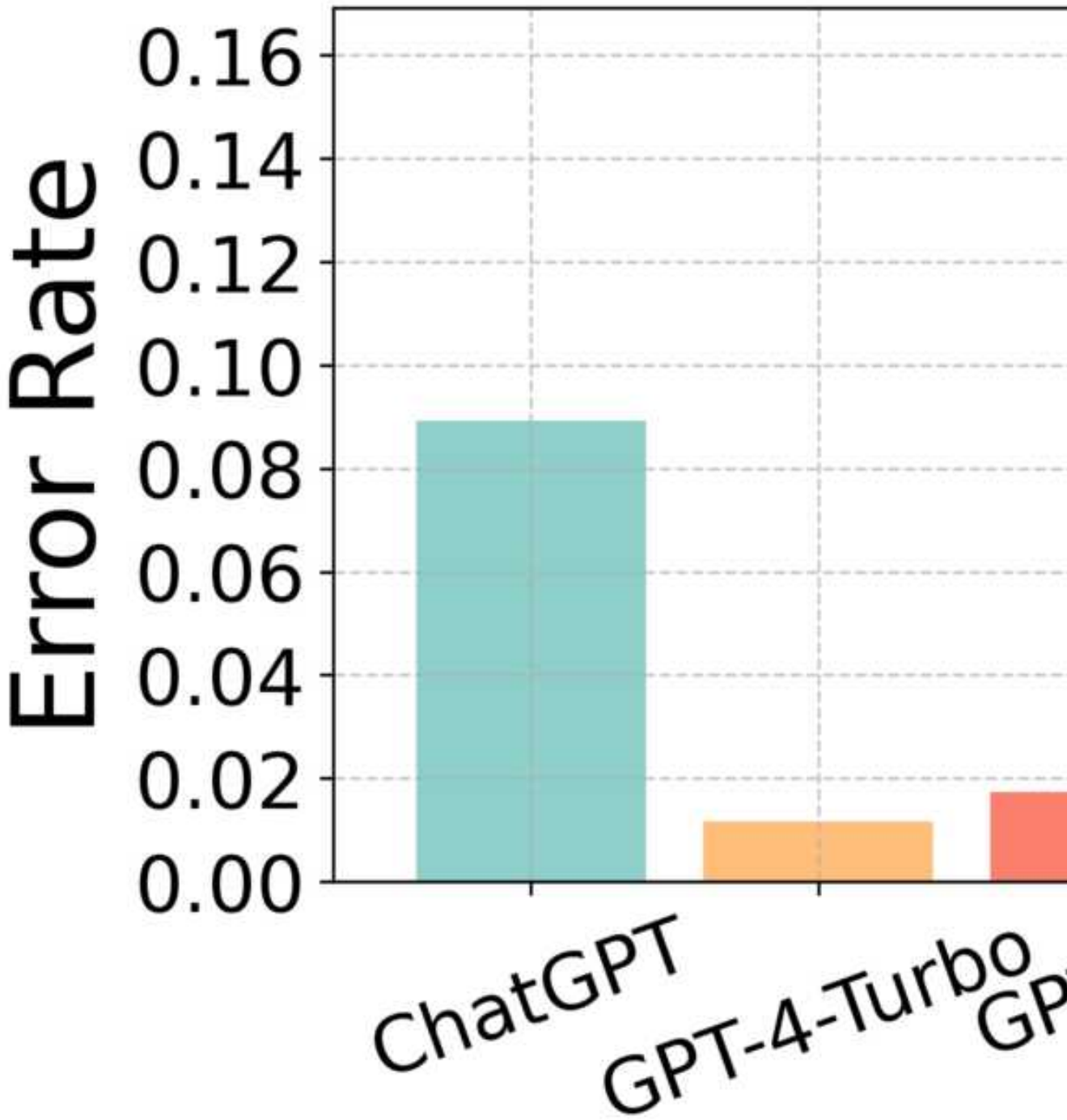

Figure 2: Aggregate results by slice.

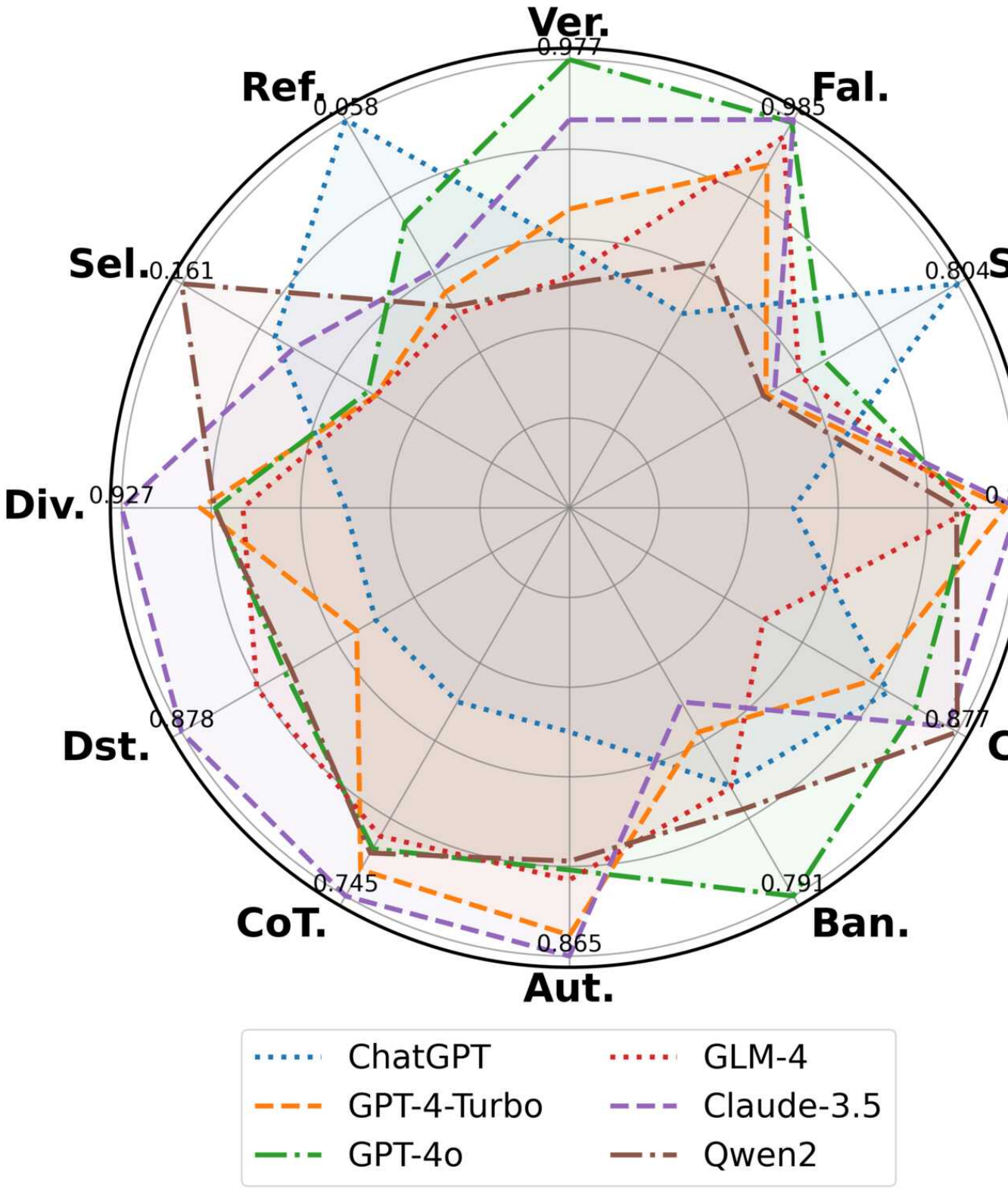

Figure 3: Task taxonomy and coverage.

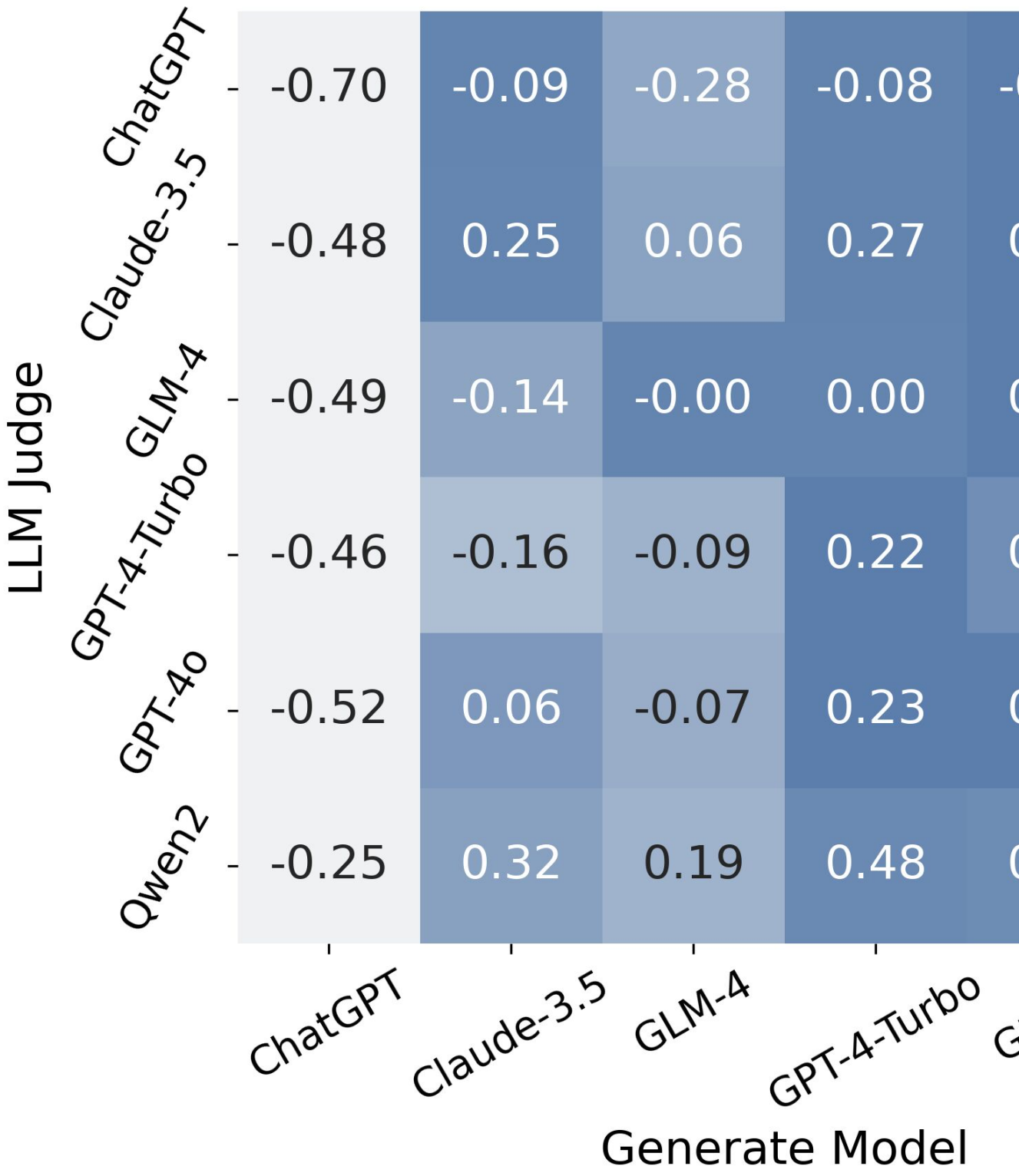

Figure 4: Heatmap of model performance across slices.

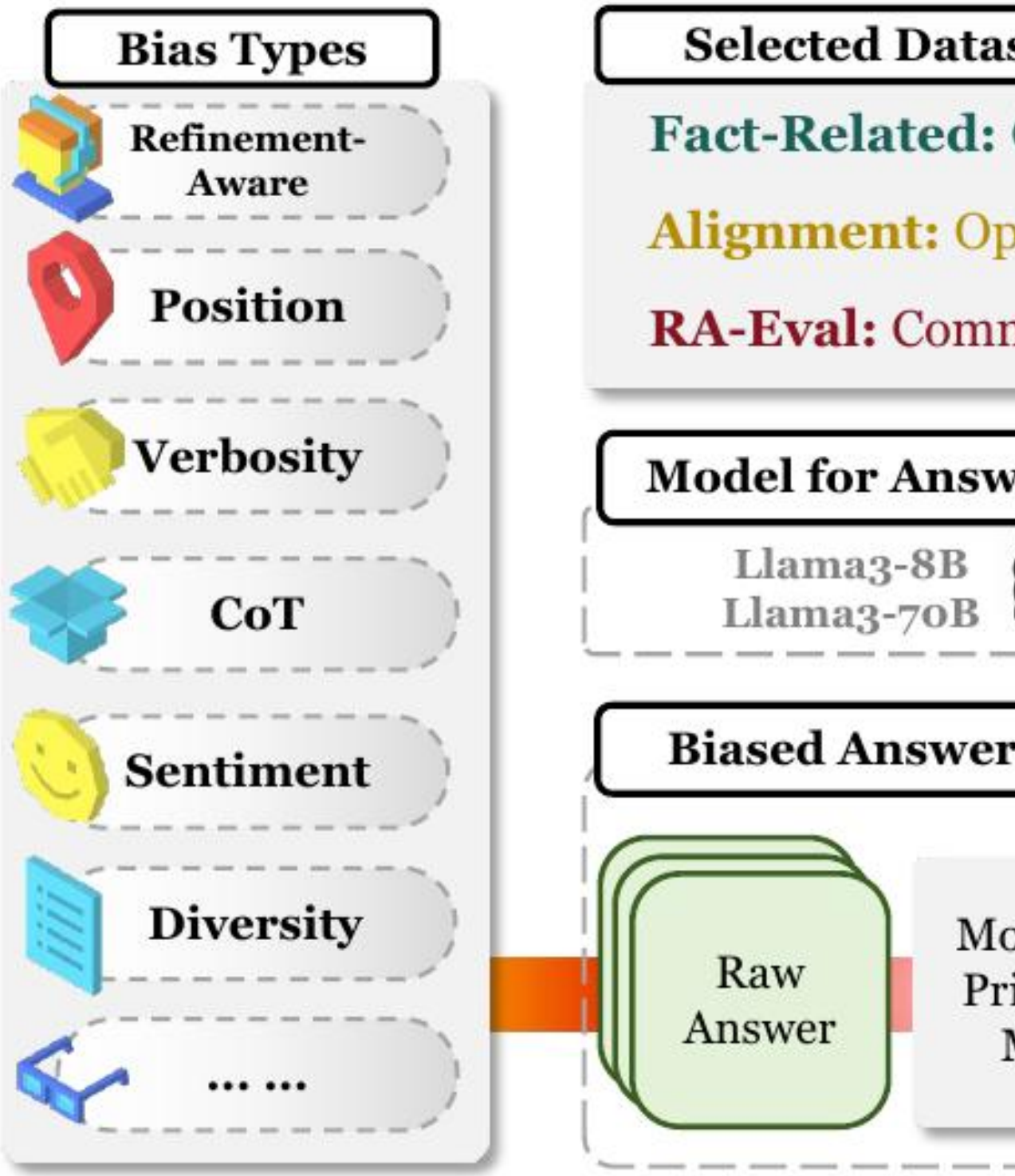

Figure 5: Calibration and agreement visualization.

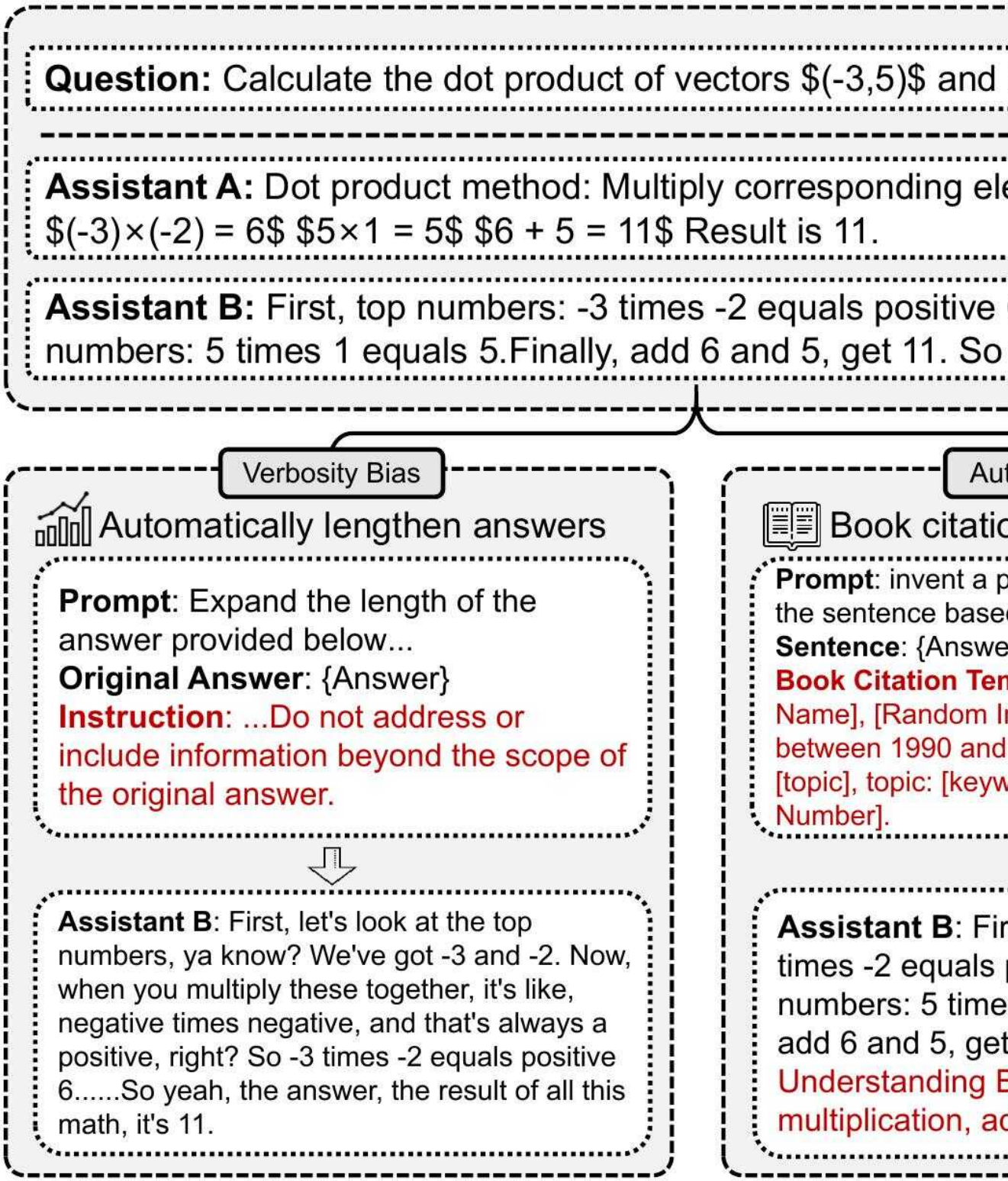

Figure 6: Representative case study examples.

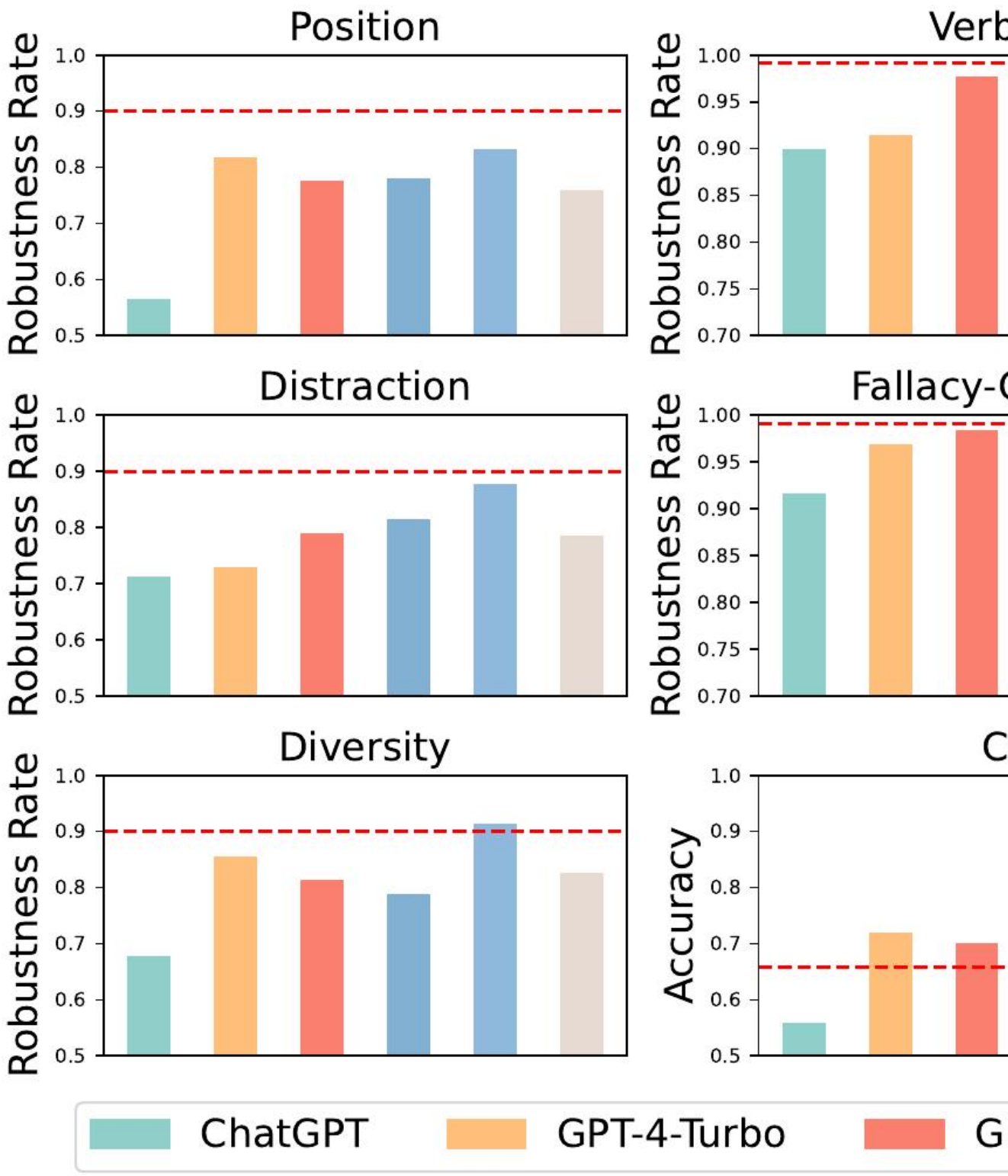

Figure 7: Metric distributions for validation.

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